

```
const API = "http://172.184.181.33/api2.php";
```

```
const tabla = document.getElementById("tabla");
```

```
const buscador = document.getElementById("buscador");
```

```
const filtro = document.getElementById("filtro");
```

```
// Crear el botón de limpiar
```

```
const limpiarBtn = document.createElement("button");
```

```
limpiarBtn.innerText = "Limpiar";
```

```
limpiarBtn.style.marginLeft = "10px";
```

```
document.querySelector(".toolbar").appendChild(limpiarBtn);
```

```
let DATA = [];
```

```
const canvas = document.getElementById("bg");
```

```
const ctx = canvas.getContext("2d");
```

```
canvas.width = innerWidth;
```

```
canvas.height = innerHeight;
```

```
let particles = Array.from({ length: 120 }, () => ({
```

```
  x: Math.random() * canvas.width,
```

```
  y: Math.random() * canvas.height,
```

```
  r: Math.random() * 2,
```

```
  dx: (Math.random() - 0.5) * 0.3,
```

```
  dy: (Math.random() - 0.5) * 0.3
```

```
}));
```

```
function animate() {
```

```
  ctx.clearRect(0, 0, canvas.width, canvas.height);
```

```
  particles.forEach((p, i) => {
```

```
    p.x += p.dx; p.y += p.dy;
```

```
    if (p.x < 0 || p.x > canvas.width) p.dx *= -1;
```

```
    if (p.y < 0 || p.y > canvas.height) p.dy *= -1;
```

```
    ctx.beginPath(); ctx.arc(p.x, p.y, p.r, 0, Math.PI * 2);
```

```
    ctx.fillStyle = "rgba(0,255,255,0.4)"; ctx.fill();
```

```

particles.slice(i + 1).forEach(p2 => {

  const d = Math.hypot(p.x - p2.x, p.y - p2.y);

  if (d < 90) {

    ctx.beginPath(); ctx.moveTo(p.x, p.y); ctx.lineTo(p2.x, p2.y);

    ctx.strokeStyle = `rgba(0,255,255,${1 - d / 90})`; ctx.stroke();

  }

});

});

requestAnimationFrame(animate);
}

animate();

function position(z) {

  const layout = {

    1: [1, 1], 2: [18, 1], 3: [1, 2], 4: [2, 2],

    5: [13, 2], 6: [14, 2], 7: [15, 2], 8: [16, 2], 9: [17, 2], 10: [18, 2],

    11: [1, 3], 12: [2, 3], 13: [13, 3], 14: [14, 3], 15: [15, 3], 16: [16, 3], 17: [17, 3], 18: [18, 3]

  };

  if (layout[z]) return layout[z];

  return [(z % 18) || 18, Math.ceil(z / 18) + 2];

}

function block(z) {

  if ([1, 3, 11, 19, 37, 55, 87].includes(z)) return "s";

  if (z >= 21 && z <= 30) return "d";

  if (z >= 57 && z <= 71) return "f";

  if (z >= 89 && z <= 103) return "f";

  return "p";

}

// Función única para estados

function getState(z) {

  if (z >= 104) return "synthetic";

  if (z <= 2) return "gas";

  if (z >= 80 && z < 104) return "liquid";

```

```
    return "solid";  
  }  
}
```

```
async function load() {  
  const res = await fetch(API);  
  DATA = await res.json();  
  render(DATA);  
}
```

```
function render(data) {  
  tabla.innerHTML = "";  
  data.forEach(el => {  
    const div = document.createElement("div");  
    const z = el.numero_atómico;  
    const [col, row] = position(z);  
    const b = block(z);  
    const state = getState(z);  
    div.className = `elemento ${b} ${state}`;  
    div.style.gridColumn = col;  
    div.style.gridRow = row;  
    div.innerHTML = `

${z}</div><div class="simbolo-estado" style="font-size:16px;font-weight:bold;">${el.simbolo}</div><div  
style="font-size:9px;opacity:.8">${el.nombre}</div>`;  
    div.onclick = () => openModal(el);  
    tabla.appendChild(div);  
  });  
}


```

```
function openModal(el) {  
  const state = getState(el.numero_atómico);  
  const modal = document.getElementById("modal");  
  modal.style.display = "flex";  
  document.getElementById("m-title").innerHTML = `${el.nombre} (${el.simbolo})`;
```

```
  let estadoTexto;
```

```
  switch(state) {
```

```

case 'liquid': estadoTexto = 'LÍQUIDO'; break;

case 'solid': estadoTexto = 'SÓLIDO'; break;

case 'synthetic': estadoTexto = 'SINTÉTICO'; break;

default: estadoTexto = 'GAS';

}

```

```

document.getElementById("m-body").innerHTML = `

<div class="data-grid">

  <div class="data-item"><b>Número atómico:</b> ${el.numero_atomico}</div>

  <div class="data-item"><b>Categoría:</b> ${el.categoria}</div>

  <div class="data-item"><b>Peso atómico:</b> ${el.peso_atomico}</div>

  <div class="data-item"><b>Descubridor:</b> ${el.descubridor}</div>

  <div class="data-item"><b>Año:</b> ${el.anio_descubrimiento}</div>

  <div class="data-item"><b>Origen:</b> ${el.origen}</div>

  <div class="data-item"><b>Familia:</b> ${el.familia}</div>

  <div class="data-item"><b>Periodo:</b> ${el.periodo}</div>

  <div class="data-item"><b>Grupo:</b> ${el.grupo ?? "N/A"}</div>

  <div class="data-item"><b>Electronegatividad:</b> ${el.electronegatividad ?? "N/A"}</div>

  <div class="data-item">Estado físico: <b class="state-${state}">${estadoTexto}</b></div>

  <div class="data-item" style="grid-column:1/3"><b>Configuración electrónica:</b><br>${el.configuracion_electronica}</div>

  <div class="data-item" style="grid-column:1/3"><b>Estado de oxidación:</b> ${el.estado_oxidacion}</div>

</div>`;

```

```

startQuantumEffect(el.numero_atomico);

modal.onclick = (e) => { if (e.target.id === "modal") closeModal(); };

document.getElementById("closeBtn").onclick = () => closeModal();

}

```

```

function closeModal() { document.getElementById("modal").style.display = "none"; }

```

```

function startQuantumEffect(Z) {

  const container = document.getElementById("orbital-container");

  container.innerHTML = "";

  const nucleus = document.createElement("div");

  nucleus.className = "quantum-core";

```

```

container.appendChild(nucleus);

const electrons = Math.min(40, Z * 2);

for (let i = 0; i < electrons; i++) {

  const e = document.createElement("div");

  e.className = "quantum-electron";

  container.appendChild(e);

  let angle = Math.random() * Math.PI * 2;

  let radius = 20 + Math.random() * 60;

  let speed = 0.01 + Math.random() * 0.02;

  function animateQuantum() {

    angle += speed;

    e.style.left = (container.clientWidth / 2 + Math.cos(angle) * radius) + "px";

    e.style.top = (container.clientHeight / 2 + Math.sin(angle) * radius) + "px";

    requestAnimationFrame(animateQuantum);

  }

  animateQuantum();

}
}

```

```

function buscarElemento() {

  const v = buscador.value.toLowerCase();

  render(DATA.filter(x =>

    x.nombre.toLowerCase().includes(v) ||

    x.categoria.toLowerCase().includes(v) ||

    x.origen.toLowerCase().includes(v) ||

    x.descubridor.toLowerCase().includes(v) ||

    String(x.anio_descubrimiento).includes(v)

  ));

}

```

```

buscador.addEventListener("input", buscarElemento);

filtro.addEventListener("change", e => {

  if (e.target.value === "all") return render(DATA);

  render(DATA.filter(x => block(x.numero_atómico) === e.target.value));

});

```

```
limpiarBtn.addEventListener("click", () => {  
    buscador.value = "";  
    filtro.value = "all";  
    render(DATA);  
});
```

```
load();
```